

10.1

$$m = 5 \text{ kg}$$

$$\epsilon_{cu} = 1,2 \cdot 10^{11} \frac{\text{N}}{\text{m}^2}$$

$$\epsilon_{jek} = 2 \cdot 10^{11} \frac{\text{N}}{\text{m}^2}$$

$$x = ?$$

(da bo palica vodoravna)



$$\left(\frac{\Delta l}{l} \right) \epsilon = \frac{F}{S}$$

se obrami

$$\frac{F_{cu}}{\epsilon_{cu} \cdot \delta} = \frac{F_{jek}}{\epsilon_{jek} \cdot \delta}$$

$$F_{cu} = \frac{\epsilon_{cu}}{\epsilon_{jek}} F_{jek}$$

$$M: F_{cu} \cdot x = F_{jek} (a - x)$$

$$\cancel{F_{jek}} \frac{a-x}{x} = \frac{\epsilon_{cu}}{\epsilon_{jek}} \cancel{F_{jek}}$$

$$a - x = x \frac{\epsilon_{cu}}{\epsilon_{jek}}$$

$$x \left(1 + \frac{\epsilon_{cu}}{\epsilon_{jek}} \right) = a$$

$$\frac{x}{a} = \frac{1}{1 + \frac{\epsilon_{cu}}{\epsilon_{jek}}}$$

10.2

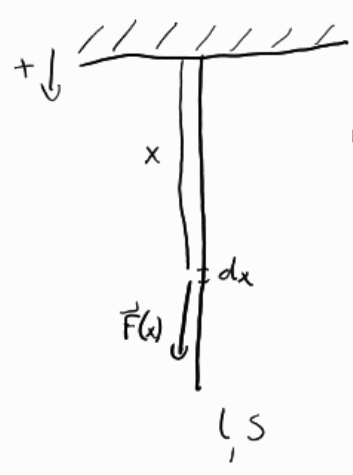
$$l = 2 \text{ m}$$

$$S = 1 \text{ cm}^2$$

$$\rho = 8900 \frac{\text{kg}}{\text{m}^3}$$

$$\epsilon = 1,2 \cdot 10^{11} \frac{\text{N}}{\text{m}^2}$$

$$\Delta l = ?$$



$$F(x) = (l - x) S \cdot \rho g$$

razsterek dx na

$$\frac{du}{dx} \epsilon = \frac{F(x)}{S} = (l - x) \cdot \rho g$$

$$\int_0^l du \cdot \epsilon = \rho g \int_0^l (l - x) dx$$

$$\Delta l = \frac{\rho g}{\epsilon} \left(lx \Big|_0^l - \frac{x^2}{2} \Big|_0^l \right) = \frac{\rho g}{\epsilon} \left(l^2 - \frac{l^2}{2} \right) = \frac{\rho g l^2}{2 \epsilon}$$

$$\Delta l = \frac{8900 \cdot 10^4 \cdot 4}{2 \cdot 1,2 \cdot 10^{11}} \text{ m} = 1,5 \cdot 10^{-6} \text{ m}$$

10.3

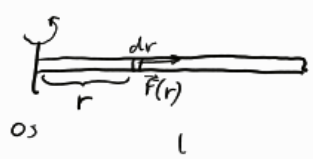
$$l = 5 \text{ m}$$

$$v = 500 \text{ min}^{-1}$$

$$\epsilon = 1,5 \cdot 10^{11} \frac{\text{N}}{\text{m}^2}$$

$$\rho = 400 \text{ kg/m}^3$$

$$\Delta l = ?$$



$$F_c = m \cdot \omega^2 r$$

$$dF_c(r) = dm \cdot \omega^2 r = \frac{m \omega^2}{l} r dr \quad dm = m \frac{dr}{l}$$

$$F_c(r) = \frac{m \omega^2}{l} \int_r^l r dr = \frac{m \omega^2}{l} \left(\frac{l^2}{2} - \frac{r^2}{2} \right)$$

$$\frac{du}{dr} \epsilon = \frac{F_c(r)}{S} = \frac{m \omega^2 (l^2 - r^2)}{2Sl}$$

$$\begin{aligned} \Delta l &= \frac{m \omega^2}{2Sl \epsilon} \int_0^l (l^2 - r^2) dr = \frac{m \omega^2}{2Sl \epsilon} \left(l^2 r - \frac{r^3}{3} \right) \Big|_0^l = \\ &= \frac{m \omega^2}{2V \epsilon} \left(l^3 - \frac{1}{3} l^3 \right) = \frac{m \omega^2 l^3}{3V \epsilon} = \frac{4 \pi^2 v^2 l^3}{3 \epsilon} = \underline{\underline{0,3 \text{ mm}}} \end{aligned}$$

10.9

$$d = 1 \text{ mm}$$

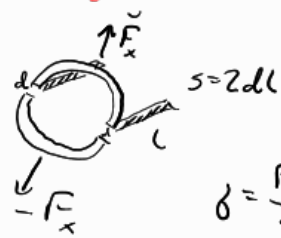
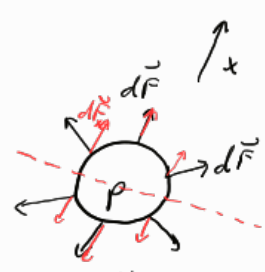
$$r = 5 \text{ cm}$$

$$\delta = 1,7 \cdot 10^8 \frac{\text{N}}{\text{m}^2}$$

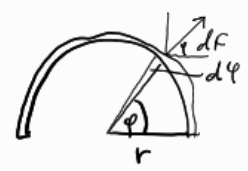
$$p_{\text{max}} = ?$$



$$p = \frac{F}{S}$$



$$\delta = \frac{F_x}{S} = \frac{F_x}{2dl}$$



$$dF = p \cdot dS = p \cdot l \cdot r d\varphi$$

$$dF_x = dF \cdot \sin \varphi$$

$$F_x = p \cdot l \cdot r \int_0^\pi \sin \varphi d\varphi = p l r (1 - (-1)) = 2 p l r$$

$$\delta = \frac{2 p l r}{2 d l}$$

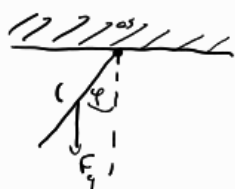
$$p = \frac{\delta d}{r} = 28 \text{ bar} = 2,8 \cdot 10^6 \frac{\text{N}}{\text{m}^2}$$

10.6

$$\Delta T = 50^\circ\text{C}$$

$$\alpha = 10^{-5} \text{ K}^{-1}$$

$$\Delta t_0 = ?$$



$$W_{p,j} = mg \frac{l}{2} (1 - \cos \varphi) = mg \frac{l}{2} \cdot \frac{\varphi^2}{2} = mg l \varphi^2 \cdot \frac{1}{4}$$

$$W_{\text{rot}} = \frac{J \omega^2}{2} = \frac{m l^2 \dot{\varphi}^2}{6} = \frac{m l^2 \dot{\varphi}^2}{6}$$

$$\frac{mg l \varphi^2}{4} + \frac{m l^2 \dot{\varphi}^2}{6} = \text{konst.} \quad / \cdot \frac{d}{dt}$$

$$\Delta l = l \cdot \alpha \cdot \Delta T$$



$$l' = l(1 + \alpha \Delta T)$$

$$t_0' = 2\pi \sqrt{\frac{2l(1 + \alpha \Delta T)}{3g}}$$

$$\alpha \Delta T \ll 1$$

$$t_0' = 2\pi \sqrt{\frac{2l}{3g}} \left(1 + \frac{1}{2} \alpha \Delta T\right)$$

$$\Delta t_0 = t_0' - t_0 = t_0 \cdot \frac{1}{2} \alpha \Delta T$$

$$\frac{mg l}{4} \cdot 2\varphi \dot{\varphi} + \frac{m l^2}{6} \cdot 2\dot{\varphi} \ddot{\varphi} = 0$$

$$g\varphi + \frac{2}{3}l\ddot{\varphi} = 0$$

$$\ddot{\varphi} = - \underbrace{\frac{3g}{2l}}_{\omega_0^2} \varphi \rightarrow t_0 = \frac{2\pi}{\omega_0} = 2\pi \sqrt{\frac{2l}{3g}}$$

2. možnost

/ln

$$\ln t_0 = \ln \left(2\pi \sqrt{\frac{2l}{3g}} \right) = \ln \left(2\pi \sqrt{\frac{2}{3g}} \right) + \ln \sqrt{l} = \ln \left(2\pi \sqrt{\frac{2}{3g}} \right) + \frac{1}{2} \ln l$$

$$\frac{dt_0}{t_0} = 0 + \frac{1}{2} \frac{dl}{l}$$

$$dt_0 = t_0 \cdot \frac{1}{2} \frac{dl}{l}$$

29/3

$$l_1 = l_2$$

$$l = 0.5 \text{ m}$$

$$S_1 = S_2$$

$$T_0 = 20^\circ\text{C}$$

$$\alpha_1 = 1.1 \cdot 10^{-5} \text{ K}^{-1}$$

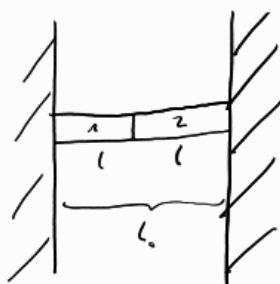
$$\alpha_2 = 1.7 \cdot 10^{-5} \text{ K}^{-1}$$

$$E_1 = 2.1 \cdot 10^5 \text{ N/mm}^2$$

$$E_2 = 9 \cdot 10^4 \text{ N/mm}^2$$

$$9.9 \cdot 10^5 \text{ N/mm}^2$$

$$\frac{F}{S} (T = 30^\circ\text{C}) = ?$$



za 1 palicu:

$$\frac{\Delta l}{l} = \alpha \Delta T \leftarrow \text{poz.}$$

$$\frac{\Delta l}{l} = \frac{F}{S} \frac{1}{E} \leftarrow \text{neg.}$$

$$\frac{\Delta l}{l} = \alpha \Delta T - \frac{F}{S} \frac{1}{E}$$

$$\Delta l_1 + \Delta l_2 = 0$$

$$l \left(\Delta T (\alpha_1 + \alpha_2) - \frac{F}{S} \left(\frac{1}{E_1} + \frac{1}{E_2} \right) \right) = 0$$

$$\frac{F}{S} \left(\frac{1}{E_1} + \frac{1}{E_2} \right) = \Delta T (\alpha_1 + \alpha_2)$$

$$\frac{F}{S} = \Delta T \frac{\alpha_1 + \alpha_2}{\frac{1}{E_1} + \frac{1}{E_2}} = \frac{10 \text{ K} \cdot 2.8 \cdot 10^{-5} \text{ N}}{10^{-5} \left(\frac{1}{2.1} + \frac{1}{9.9} \right) \text{ K mm}^2} = \underline{\underline{176 \frac{\text{N}}{\text{mm}^2}}}$$

29/7

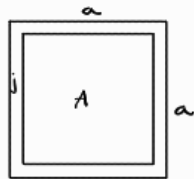
$$\Delta p = 6 \text{ bar}$$

$$\chi_A = 1,15 \cdot 10^{-4} \text{ bar}^{-1}$$

$$\beta_A = 7,5 \cdot 10^{-4} \text{ K}^{-1}$$

$$\alpha_j = 1,2 \cdot 10^{-5} \text{ K}^{-1}$$

$$\Delta T = ?$$



alkohol:

$$\frac{\Delta V}{V} = \beta \Delta T$$

$$\frac{\Delta V}{V} = -\chi \cdot \Delta p$$

$$\rightarrow \boxed{\frac{\Delta V}{V} = \beta \Delta T - \chi \Delta p}$$

jeklena posoda:

$$V = a^3$$

$$V + \Delta V = (a + \Delta a)^3 = a^3 + 3a^2 \Delta a + 3a(\Delta a)^2 + (\Delta a)^3$$

$$\Delta V = 3a^2 \Delta a + a(\Delta a)^2 + (\Delta a)^3$$

$$\Delta V = 3a^2 \Delta a$$

zanemarimo, ker Δa majhen

$$\Delta a = a \alpha_j \Delta T$$

$$\Delta V = 3a^3 \alpha_j \Delta T = 3 \cdot V \cdot \alpha_j \cdot \Delta T$$

$$\boxed{\frac{\Delta V}{V} = 3 \alpha_j \Delta T}$$

Lažje = diferenciali:

$$V = a^3$$

$$\ln V = \ln a^3 = 3 \ln a$$

$$\frac{dV}{V} = 3 \cdot \frac{da}{a} = 3 \cdot \alpha_j dT$$

$$\boxed{\frac{dV}{V} = 3 \alpha_j dT}$$

$$3 \alpha_j \Delta T = \beta_A \Delta T - \chi \Delta p$$

$$\Delta T (\beta_A - 3 \alpha_j) = \chi \Delta p$$

$$\Delta T = \frac{\chi \Delta p}{\beta_A - 3 \alpha_j} = 1 \text{ K}$$